Anvesh Reddy

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EDUCATION

University of Illinois at Urbana-Champaign *Expected Graduation: Dec 2027*
Grainger College of Engineering | B.S. in Engineering Mechanics (Concentration in Robotics) **GPA: 3.73/4.00**
Grainger College of Engineering | Minor in Computer Science

TECHNICAL SKILLS

Coursework: Statics, Dynamics, Solid Mechanics, Thermodynamics, Design for Manufacturability & Introduction to Robotics
CAD & Design: Fusion 360, SolidWorks, Onshape, Siemens NX, Apriori cost analysis, GD&T, DFMA
Programming: Python, C++, Kotlin, Git, MATLAB, ROS, Claude, Gemini Anti-Gravity

WORK EXPERIENCE

Technology Entrepreneur Center – Champaign, IL *January 2025 - Present*
Advisor and Teaching Assistant
- Organized 10+ pitch workshops and demo-day for 250+ teams for my university's premier startup competition: COZAD
- Graded 30+ pitch decks for technology, engineering, and healthcare startups providing strategic feedback on go-to-market plans that helped them secure significant seed funding.

Technology Services Helpdesk – Champaign, IL *October 2025 – Present*
Team Lead
- Leading a team of 50 helpdesk consultants, overseeing ticket assignments and scheduling logistics across 3 locations on campus to maintain a customer satisfaction rating of 97.7%
- Diagnosed and resolved 250+ technical tickets and 100+ support calls for complex hardware, software, and network failures for university students and faculty as a helpdesk consultant.

TECHNICAL PROJECTS

UR3 Robotic Arm Programming – Champaign, IL *January 2026 – April 2026*
- Implemented forward and inverse kinematics for a 6-DOF UR3 robotic arm in ROS/Python, deriving all six joint angle solutions from a desired end-effector position and gripper orientation in world-frame coordinates.
- Programmed a Tower of Hanoi pick-and-place algorithm on the UR3 robotic arm in ROS/Python, autonomously sequencing disk moves by integrating inverse kinematics and suction gripper control to solve the puzzle without human intervention

DC Motor Housing Design & Integration – Champaign, IL *August 2025 – December 2025*
- Designed and iterated a custom ABS housing in Fusion 360 featuring tool-free assembly via snap-fit joints and internal ribbing, optimizing ergonomics and a sub-85g weight target while maintaining structural integrity under loading from the DC motor.
- Performed root-cause analysis on component tolerances using digital calipers and feeler gauges, identifying a 0.45mm deviation in motor contact slots, and redesigning the interface geometry to achieve consistent press-fit compliance.
- Applied Cradle-to-Cradle design principles to repurpose a DC motor and drive assembly from a consumer product, extending component lifecycle through Design for Disassembly and reducing material waste across the full product architecture.

Modular Trash Pick Design & DFM Optimization – Champaign, IL *January 2025 – May 2025*
- Created detailed engineering part drawings and oversaw the assembly of 8 interdependent components, ensuring dimensional compliance with ASME Y14.5 (GD&T) specifications across all technical documentation.
- Executed 3+ design iterations of the handle assembly by integrating user interview feedback, applying ergonomic principles and rapid prototyping to improve grip comfort and reduce operator hand fatigue.
- Optimized part geometry for high-volume injection molding using Apriori cost analysis, achieving a 20% reduction in manufacturing costs while maintaining all functional and tolerance requirements

LEADERSHIP AND PROFESSIONAL DEVELOPMENT

Phi Gamma Nu Professional Business Fraternity – Champaign, IL *February 2026 – Present*
- Selected as 1 of 23 from a pool of 800+ applicants, participating in 30+ professional development workshops focused on leadership, communication, and business acumen.

Silicon Valley Entrepreneurship Workshop – Bay Area, CA *December 2025 – January 2026*
- Selected as 1 of 25 from a pool of 400+ applicants to visit 23 startups founded by UIUC alumni, gaining firsthand exposure to product development lifecycles, rapid prototyping workflows, and cross-functional engineering team operations.

Grainger Global Sustainability Scholar's Program – Trinidad and Tobago *December 2024 – January 2025*
- Assisted Fondes Amandes in installing a solar-powered 360-degree camera on their fire tower and setting up an on-site weather monitoring station to improve visibility and environmental data collection in the forests of Trinidad.
- Compiled a grant application database and sent outreach emails to donors to help ERIC acquire funding for a pilot test.